

Relativistic and Spinor Witnesses

Volume 0. Exact conditional correspondences between the half-angle calculus and accepted relativistic, weak, Coulomb, and electromagnetic structures — and the audit that keeps them honest.

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About this book. Rebuilt as standalone R Theory from the former T4 essay-book. It asks one question of each imported physical structure: does the canonical half-angle calculus occur inside it exactly, and if so, does anything new follow? The answer, section by section, is *yes, exactly* — *and no, nothing new*. That second clause is not a disappointment; it is the result. Every import is declared, every obstruction is kept, and the book closes with a formal prediction-difference audit.

Status key. **EXACT** proved mathematics · **CONDITIONAL** exact after a declared import or contract · **IMPORT** accepted physics taken as input, not derived · **OBSTRUCTION** a proved or retained negative result.

Depends on: [Book 0](#) (primitive calculus, one-generator reduction); the internal paper *Reciprocal Half-Angle Coordinates in the Radial Dirac–Coulomb Problem*.

§1 Kinematics

§2 Spinor

§3 V–A

§4 Beta/Coulomb

§5 Electromagnetism

§6 Audit

§1 — The relativistic kinematic witness

Take a massive particle moving at speed β . There is a classical angle hiding in its kinematics: the angle χ with $\sin \chi = \beta$ and $\cos \chi = mc^2/E$. On this chart the half-angle coordinate $q = \text{sxp}(\chi) = \tan(\chi/2)$ is simultaneously three familiar things: the free-Dirac lower/upper spinor component ratio, the rational generator of β and γ , and the square root of the light-cone ratio. The reciprocal pair $\text{cxp} = e^{\eta}$, $\text{crx} = e^{-\eta}$ is the rapidity itself — and $\text{cxp} \cdot \text{crx} = 1$ is exactly the normalized mass shell, already imposed by the reference theory.

Kinematic dictionary **CONDITIONAL**

Under the contract $\sin \chi = \beta$, $\cos \chi = mc^2/E$ on the positive-energy branch:

$$\text{sxp}(\chi) = pc/(E+mc^2) = \tan(\chi/2), \quad \text{srx}(\chi) = (E+mc^2)/pc$$

$$\text{cxp}(\chi) = (E+pc)/mc^2 = e^\eta, \quad \text{crx}(\chi) = (E-pc)/mc^2 = e^{-\eta}$$

The mapping is invertible on the declared branch; the reciprocal identity is the mass shell. This is a coordinate theorem after importing special-relativistic kinematics and the free Dirac spinor —

IMPORT both.

OBSTRUCTION **Rank is not created by notation.** Evaluating many functions on the single coordinate χ does not manufacture additional independent physical degrees of freedom. The dictionary above is one angle's worth of content, however many names it carries.

§2 — Spinor completion and the native ellipse

One real coordinate is not a spinor. Lifting $\mathbf{q} = \text{sxp}(\chi)$ to a complete two-state spinor requires an **independent relative phase** ϕ — the one-generator obstruction survives physical completion. The completed state has an exact Bloch vector $(\cos \chi \cos 2\phi, \cos \chi \sin 2\phi, \sin \chi)$, and the reciprocal pair becomes exact component-amplitude odds. The native complex carrier $\Psi = H + iV$ is characterized exactly — an anisotropic real-linear squeeze of the circular phase with the counter-rotating decomposition $\Psi = (i/8)(e^{2ix} + 3e^{-2ix})$ — but without an independently selected physical embedding of its quadratures it determines no unique observable.

Spinor completion **CONDITIONAL**

With independent phase ϕ , the completed spinor is an exact coordinate representation of imported two-state geometry. No new Hamiltonian, interaction, phase law, or observable is supplied. The bare native carrier as unspecified physical embedding is **Class B** — its prediction set is undefined until a complete contract and dynamics are supplied.

§3 — Weak V–A helicity weights and the chirality-orientation

obstruction

Now import the Standard Model's left-chiral charged-current projector $P_L = (1 - \gamma^5)/2$ — imported, not derived. For a massive electron the favored/wrong helicity probabilities are $(1 \pm \beta)/2$, and their amplitude odds are exactly the reciprocal pair:

$$\sqrt{(P_{\text{fav}}/P_{\text{wrong}})} = \text{cxp}, \quad \sqrt{(P_{\text{wrong}}/P_{\text{fav}})} = \text{crx}$$

The equality is exact under the §1 kinematic contract. But the calculus is left–right symmetric: the mirror right-chiral orientation is equally available to the internal reciprocal structure until the Standard Model orientation is imported. Projection reconstructs the helicity-weight geometry of an imported chiral interaction; it does not derive why the interaction is left-chiral. That is the **chirality-orientation obstruction**, and it is retained, not repaired.

CONDITIONAL **V–A helicity correspondence.** After importing the left-chiral orientation, cxp/crx are the exact favored/wrong helicity amplitude odds. **OBSTRUCTION** The orientation itself is not derived; a Projection-only derivation of chirality selection is Class B (unspecified).

§4 — Beta decay, Coulomb dressing, and the two-angle bridge

Follow the emitted beta electron into the daughter nucleus's Coulomb field. Two independent physical angles are compulsory — not one. The continuum kinematic angle χ ($\sin \chi = \beta$, from §1) encodes the electron's mass-shell kinematics; the Coulomb structural angle a ($\sin a = Z\alpha$, $\cos a = \sqrt{1 - (Z\alpha)^2}$) encodes the daughter's Coulomb coupling. Identifying them would force $\beta = Z\alpha$ and collapse the continuous beta spectrum to a charge-selected velocity — the **two-angle necessity theorem**. The standard Coulomb parameter becomes their exact quotient,

$$\xi = Z\alpha/\beta = \sin a / \sin \chi = u(1+q^2) / [q(1+u^2)]$$

with $u = \text{sxp}(a)$, $q = \text{sxp}(\chi)$. The baseline allowed phase-space measure becomes fully rational in q and the endpoint coordinate q_0 . But the relativistic Fermi function keeps its non-elementary core $\exp(\pi\xi) |\Gamma(\lambda + i\xi)|^2$: the half-angle substitution rationalizes the kinematic and projective layers but

does not generate the complex-Gamma Coulomb scattering dynamics. The structural angle a is the $K=1$ member of the Coulomb circle of the radial Dirac–Coulomb paper — the bound and continuum problems share the chart but use it differently ($E_{\text{bound}}/m = \cos a < 1$ versus $E_{\text{cont}}/m = \sec \chi > 1$).

Two-angle Coulomb structure CONDITIONAL

- a and χ are generically independent; $a = \chi$ would incorrectly force $\beta = Z\alpha$.
- $\xi = \sin a / \sin \chi$ exactly; phase space rational in (q, q_0) .
- The Fermi function, nuclear matrix elements, radiative and finite-size corrections, endpoint energy, and the nuclear scale R remain IMPORT s.
- OBSTRUCTION The complex-Gamma factor lies outside the finite rational primitive closure — the special-function layer is imported dynamics, not derived.

§5 — Electromagnetic comparison: the adversarial audit

This section is deliberately adversarial. The question is not whether half-angle, Cayley, reciprocal, or doubled-angle forms occur in electromagnetism — they do, abundantly. The question is whether any occurrence survives the redundancies of electromagnetic description. The audit walks through nine witnesses and strips each one:

- **Jones polarization:** $z = e^{i\phi} \exp(\alpha)$ exactly — but phase ϕ and intensity scale A^2 are independent imported data (coherency/Stokes).
- **Duality:** $2V = \cos 2x$, $4H = \sin 2x$ give an exact $SO(2)$ duality-rotation matrix — but constant duality is an accepted symmetry, not a new prediction.
- **Native carrier:** $\Psi = H + iV$ is an anisotropic squeeze with exact 1:3 counter-rotating decomposition — but direct $H \leftrightarrow E$, $V \leftrightarrow B$ assignment fails an ordinary traveling wave; the **potential-first** contract $A \propto H$ recovers the standard wave after differentiation (a repair that reproduces Maxwell, not a new law).
- **Local duality:** making the duality angle spacetime-dependent gives an $SO(2)$ connection with zero curvature — pure gauge. The scalar alone does not defeat C_0 by being made local.

- **Newman–Penrose:** cpx/crx match reciprocal boost weights exactly — but those are tetrad-gauge weights, not new observables.
- **Null Maxwell spinor:** the degree-two Veronese map $[1:w:w^2]$ — but only the null rank-one sector, not the generic field.
- **Two CP^1 factors:** polarization CP^1 and propagation-direction CP^1 are independent state spaces; one real scalar cannot supply both.
- **Conformal dressing:** in four dimensions the vacuum Maxwell equations are conformally invariant on 2-forms — a scalar conformal factor changes nothing in source-free propagation.
- **Helicity:** electromagnetic helicity (massless spin-1) is not weak chirality (massive Dirac); the $q=1/2$ numerical coincidence (odds 3:1 in both sectors) promotes no cross-sector law.

The repeated verdict: exact coordinate correspondences, no surviving Projection-specific invariant.

OBSTRUCTION **Invariant-survival audit.** After removing projective, gauge, tetrad, duality, and conformal redundancies, every electromagnetic witness above is an exact correspondence inside known representation freedom. None establishes a Projection-specific invariant. C0 is strengthened, not weakened, by the redundancy analysis.

§6 — The prediction-difference audit and book closure

Correspondence is not promotion. For a fully specified Projection-augmented model MP against its accepted reference M0, define the operational prediction-difference set

$$\Delta_{\text{op}}(\text{MP}, \text{M0}) = \{ c \in C_{\text{exp}} : P_P(\cdot | c) \neq P_{\text{0}}(\cdot | c) \}$$

— the experimental configurations where the predictions actually differ. Three terminal statuses: **Class A** (fully specified, $\Delta_{\text{op}} = \emptyset$), **Class B** (incomplete — Δ_{op} *undefined*, and an open ansatz is not a nonempty prediction set), **Class C** (fully specified, $\Delta_{\text{op}} \neq \emptyset$). Two firewall theorems: an invertible representation change can never create a nonempty Δ_{op} , and incompleteness is not novelty.

The audit walks every construction of §§1–5. Each fully specified one — the kinematic dictionary, the completed spinor, the V–A odds, the two-angle beta rewrite, every electromagnetic witness — lands in **Class A**. Each potentially nonredundant but unfinished proposal — bare native-carrier embedding,

Projection-only chirality selection, scalar-only local duality dynamics — lands in **Class B**. **No Class-C candidate is present.**

Candidate status matrix (condensed)

Relativistic primitive dictionary	Class A, $\Delta=\emptyset$
Free-Dirac half-angle witness	Class A, $\Delta=\emptyset$
Completed two-state spinor	Class A, $\Delta=\emptyset$
Native ellipse (bare mathematics)	exact internal theorem
Native ellipse (physical embedding) ..	Class B, Δ undefined
V–A helicity-weight correspondence ...	Class A, $\Delta=\emptyset$
Projection-only chirality selection ..	Class B, Δ undefined
Beta/Coulomb two-angle rewrite	Class A, $\Delta=\emptyset$
Jones/coherency witness	Class A, $\Delta=\emptyset$
Constant Maxwell duality	Class A, $\Delta=\emptyset$
Potential-first Maxwell wave	Class A, $\Delta=\emptyset$
Scalar-only local duality	pure-gauge Class A / failed symmetry
NP reciprocal boost weights	Class A, $\Delta=\emptyset$
Null Maxwell spinor / Veronese	Class A, $\Delta=\emptyset$
Conformal scalar dressing	Class A, $\Delta=\emptyset$
Class C candidates	none

Book 20 closure **EXACT**

The canonical calculus repeatedly furnishes exact, unusually compact adapted coordinates on standard relativistic, spinorial, weak-helicity, Coulomb, and electromagnetic structures. By explicit obstruction and operational audit, the presently specified correspondences do not alter the invariant predictions of their accepted reference theories. Accordingly **C0 — Projection Craft changes representation but not physics** — is retained as the undefeated null hypothesis, and $\Delta_{\text{op}} = \emptyset$ for every fully specified construction in this book. This is a theorem about the content of Book 20, not a universal no-go: C0 remains falsifiable by a future fully specified Class-C model with a stated measurement contract.

Strongest positive results retained: the one-contract kinematic dictionary ($\text{sxp} \leftrightarrow \text{Dirac ratio}$, $\text{cxp} \leftrightarrow e^\wedge \eta$, $\text{cxp} \cdot \text{crx} = 1$ as mass shell); the independent-phase spinor completion; V–A odds as cxp/crx after chiral import; the two-angle beta/Coulomb structure with $\xi = \sin a/\sin x$; rational beta phase space; the exact native-carrier characterization. **Strongest negative results retained:** one scalar gains no rank

from many evaluations; chirality orientation not derived; $a \neq \chi$ necessarily; no complex-Gamma from rational closure; direct E/B assignment fails; local scalar duality is pure gauge; NP weights are tetrad-gauge; conformal dressing invisible to source-free Maxwell; helicity sectors not identified by coincidence. Later work must survive these obstructions rather than restart before them.

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A Volume 0 book of the R Theory rewrite. Source: the former T4 essay-book, rebuilt standalone. Status discipline: exact / conditional / import / obstruction — nothing promoted beyond its support. $\Delta_{\text{op}} = \emptyset$ throughout.